Epidemiology of MDROs Across 3 Veterans Affairs Nursing Homes and MDRO Transmission During Interactive Visits

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Background

- Veterans Affairs (VA) nursing homes (NHs) are often co-located within acute-care hospitals
- NH patients routinely have interactive visits during their stay, such as therapy, dialysis, or radiology
- We characterize multidrug-resistant organism (MDRO) transmission dynamics involving patients, healthcare providers, and the environment

Methods

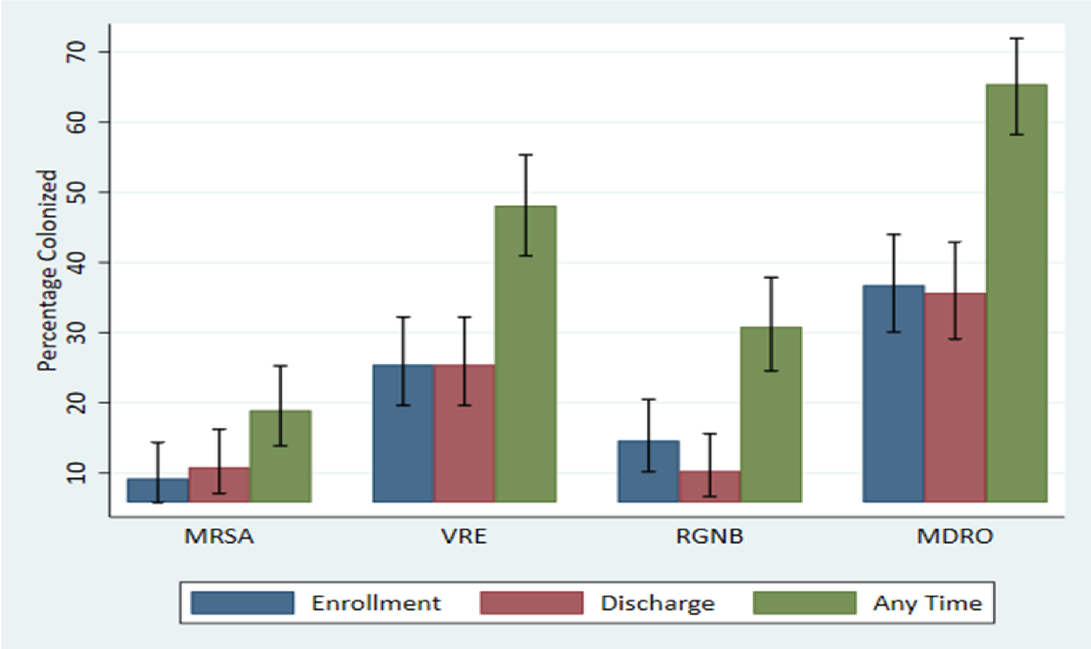
- Newly admitted Veterans at 3 VA NHs (April '21-Sept '23)
- Participants visited on day 0, 7, 14, 21, 30, 60, and 90. Samples collected from nares, groin, hands, and room surfaces, plus surfaces and hands at interactive visits (e.g., therapy, dialysis)
- Tested for methicillin-resistant *Staphylococcus aureus* (MRSA), vancomycin-resistant *Enterococcus* (VRE), and resistant gram-negative bacteria (R-GNB)

Results

197 patients enrolled (avg. 70 years; 96% male; 64% Caucasian; 33% Afr. American; median follow-up 33 days)

- **758 follow-up visits** (mean 3.8/patient; 7,543 swabs)
- **345 interactive visits** (n=135 patients; mean 2.6/patient; 3,002 swabs)

Figure 1. MDRO colonization on enrollment, at discharge, and any time during the study



37% of patients were colonized at baseline; 64% of patients were colonized at any time.

Figure 2. MDRO colonization at sampled body sites, by length of preadmission hospitalization

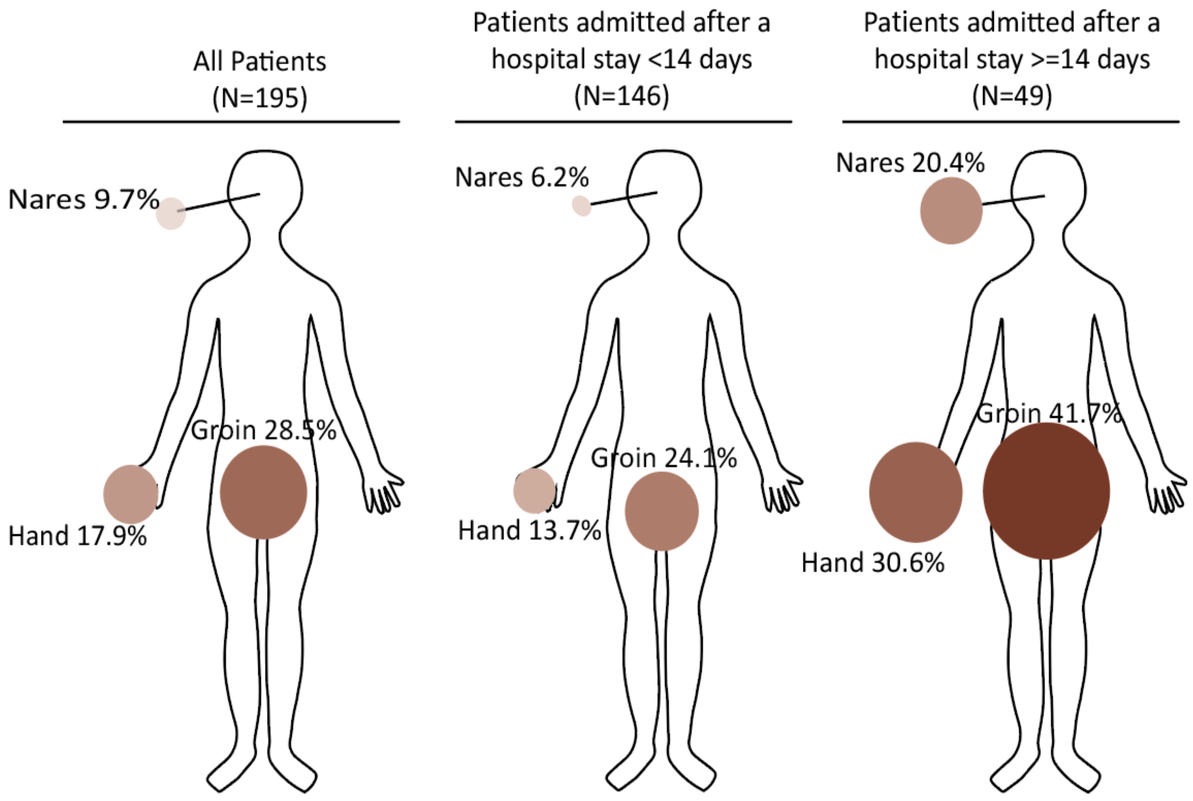


Figure 3. Transmission of VRE at an interactive visit

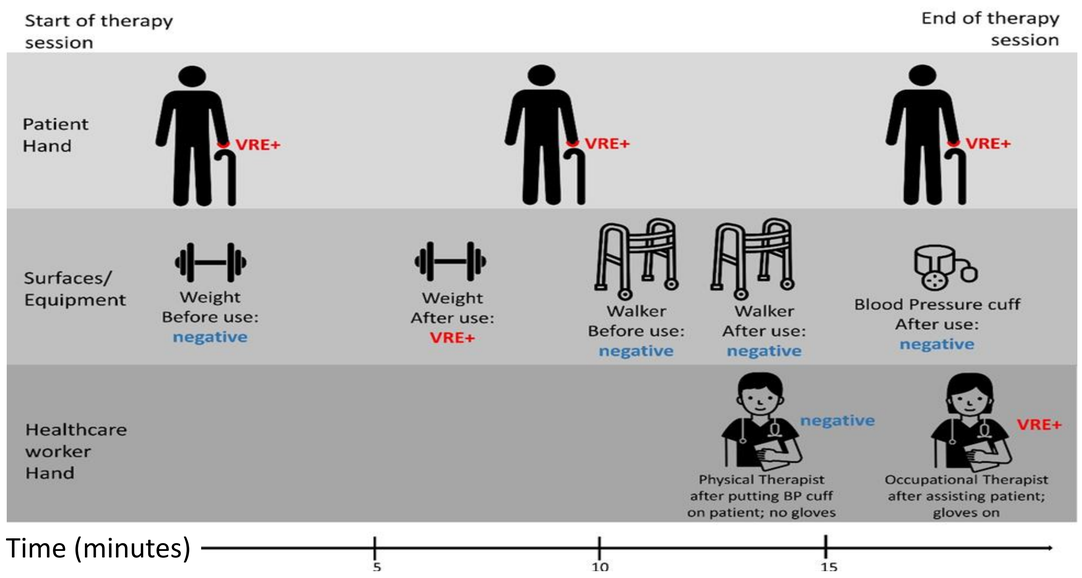
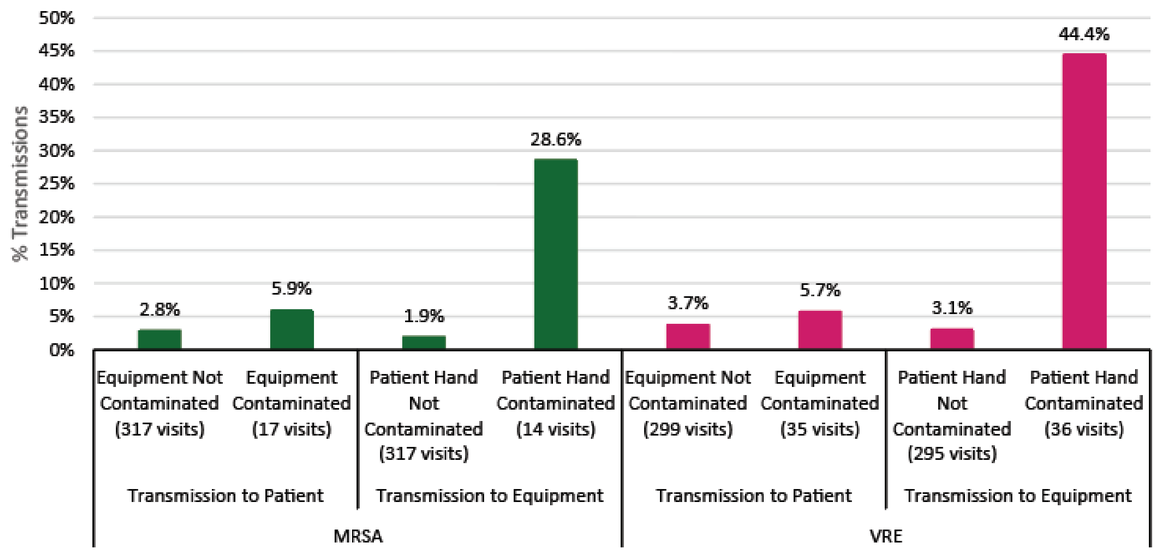


Figure 4. Contamination preceding transmissions during interactive visits



- Prevalence and new acquisition of MDROs in NH patients was very high
- Patient hand contamination was common, particularly in those with prolonged hospitalization resulting in MDRO transmission during interactive visits
- Interventions to reduce this transmission are urgently needed

Limitations: Uneven recruitment and visit frequencies as a result of COVID-19; several bacteria species and resistance patterns grouped into R-GNB category.

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